

Coupling Kratos Multiphysics with OpenFOAM for Strongly Coupled FSI Problems

Eric Sastre i Rienitz

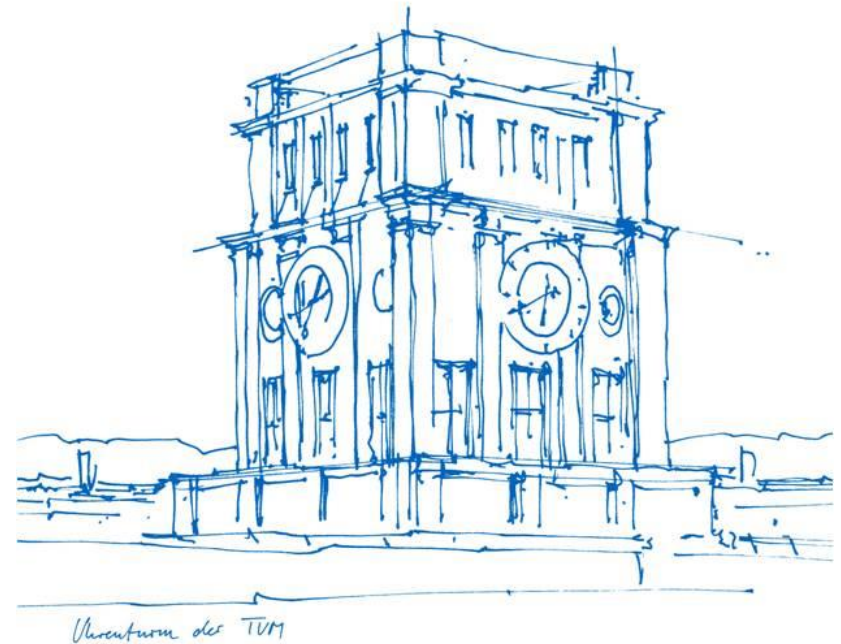
Supervisors: M.Sc. Juan Ignacio Camarotti and
M.Sc. Tobias Siemer

Technische Universität München

TUM School of Engineering and Design

Lehrstuhl für Statik und Dynamik und Lehrstuhl für
Strukturauslegung in der Luft- und Raumfahrt

Munich, 17. August 2026



Presentation Outline

- Partitioned approach: Kratos + OpenFOAM
- Previous Work: Weak-Coupling Adapter
- Developed Approach: Strong-Coupling Adapter
- Benchmark Cases
 - Mok
 - Turek
 - 3D Wing

Partitioned approach: Kratos + OpenFOAM

Partitioned approach: Kratos + OpenFOAM

- **Co-simulation** is a widely used approach for the numerical simulation of **fluid-structure interaction (FSI)** problems.
- The **partitioned approach** is the most widely used because it allows leveraging **existing mature single-field solvers of different fidelities**



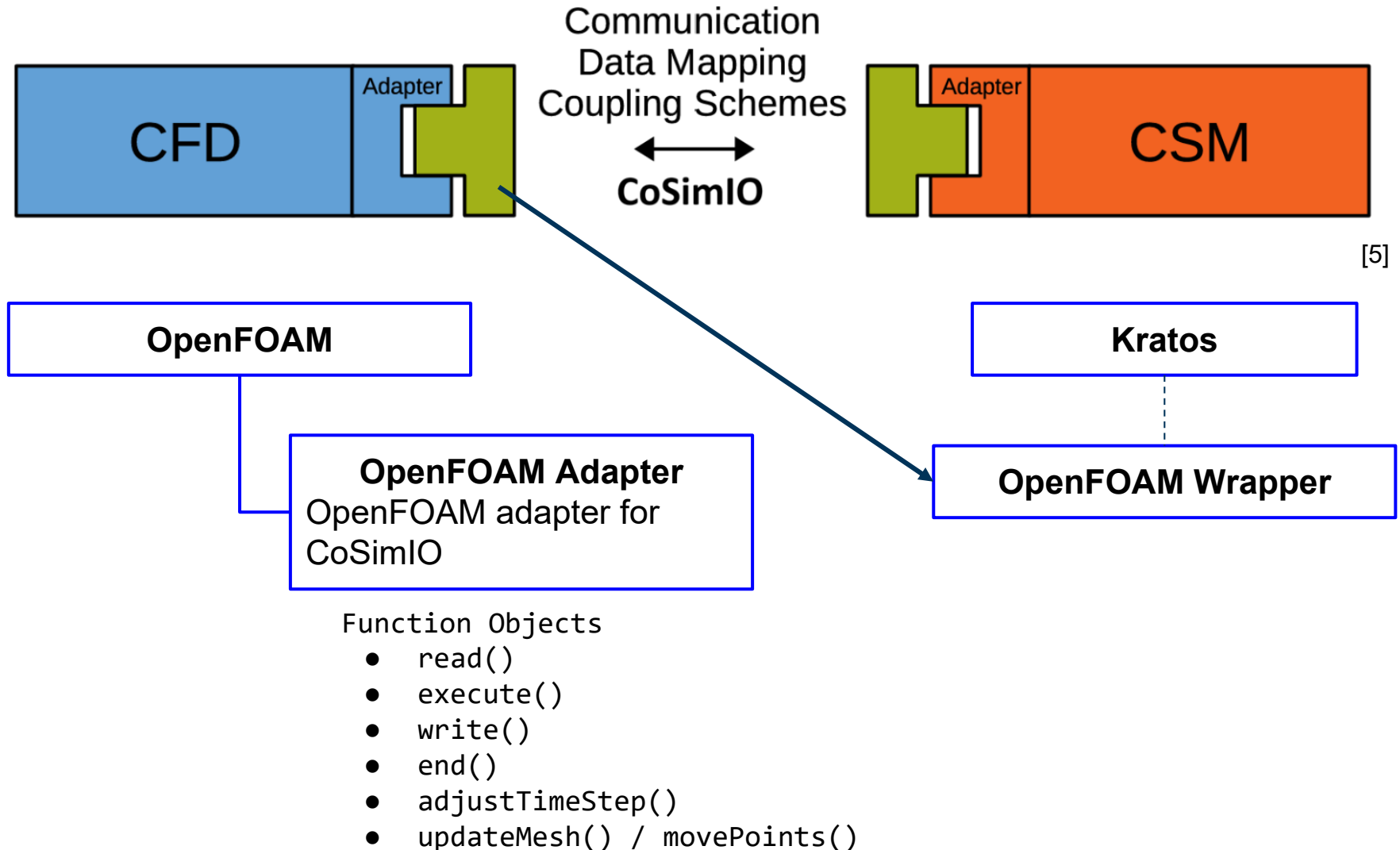
OpenFOAM is a **free, open-source CFD software** widely used in engineering and science for simulating **complex fluid flows** and **multiphysics problems**. It is professionally developed and rigorously tested for performance, reliability, and scalability.



KRATOS Multiphysics (Kratos) is a **high-performance, modular, and extensible** framework for developing **parallel, multidisciplinary simulation software**. It is written in C++ and provides an extensive Python interface.

- CoSimIO

Partitioned approach: Kratos + OpenFOAM



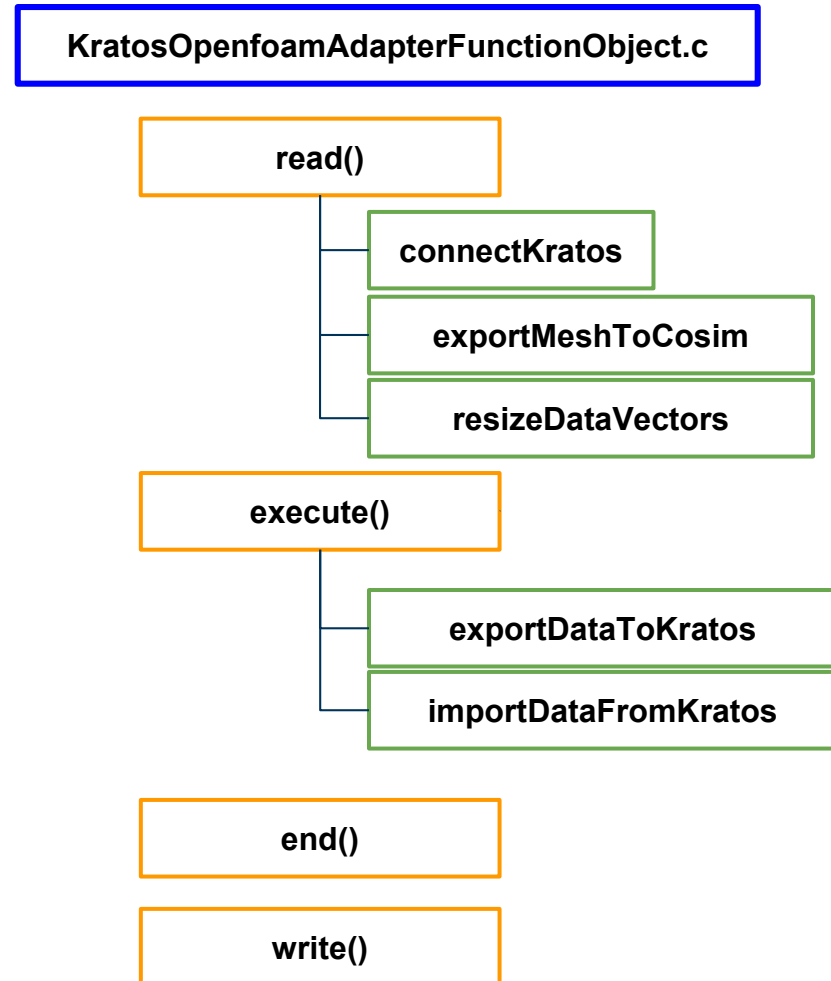
Previous Work: Weak-Coupling Adapter

Previous Work: Weak-Coupling Adapter

Ashish Shankar Darekar [6]

“Development of a General OpenFOAM-Coupling-Adapter for the KratosCoSimulation Multiphysics Solver”.

Master’s Thesis.
Technical University of Munich, 2021.



Previous Work: Weak-Coupling Adapter

```
functions
{
    // Function object "KratosOpenfoamAdapterFunctionObject" to couple OpenFOAM with CoSimulation (kratos)
    KratosOpenfoamAdapterFunctionObject1
    {
        type KratosOpenfoamAdapterFunctionObject;
        libs ("../KratosOpenfoamAdapterFunctionObject/libKratosOpenfoamAdapterFunctionObjectFunctionObjects.so");

        interfaces
        {
            Interface1
            {
                name                interface_flap;
                patches              (flap);
                importDataIdentifier (disp_interface_flap);
                exportDataIdentifier (load_interface_flap);
            }
        }

        parameters
        {
            rho rho [1 -3 0 0 0 0 0] 0.00118;
            nu  nu [0 2 -1 0 0 0 0] 0.1542;
        }

        dim 3;
        debugLevel info;
    }
}
```


Developed Approach: Strong-Coupling Adapter

Developed Approach: Strong-Coupling Adapter



A **preCICE** adapter (**function object**) for **OpenFOAM** enabling extensible surface and volume coupling for **FSI**, conjugate heat transfer, and fluid–fluid simulations.



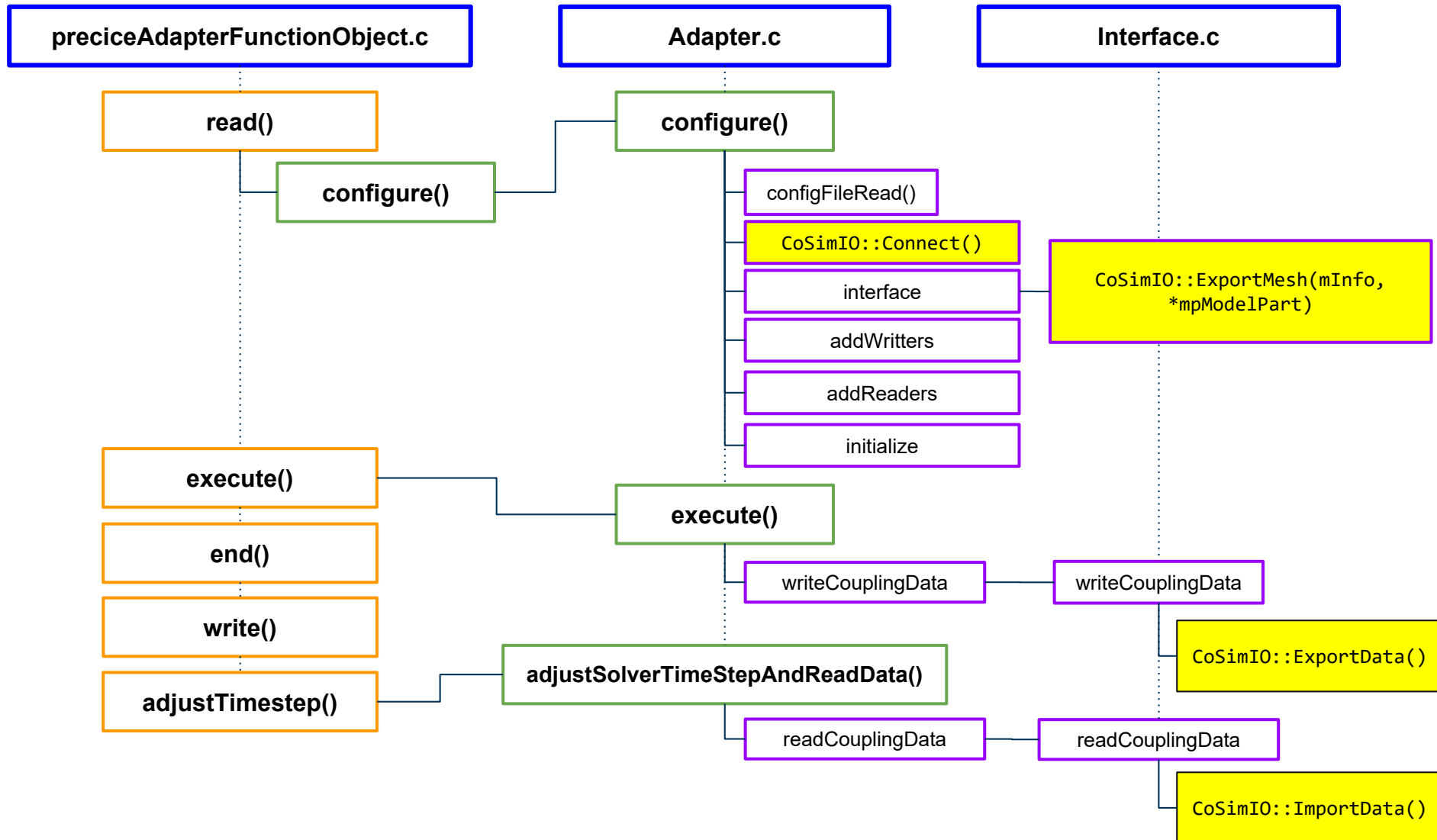
CoSimIO

Developed Approach: Strong-Coupling Adapter

 .github	Bump actions/upload-artifact from 6 to 7 (precice#399)	last month	 Allclean	More robust building scripts (precice#157)	5 years ago
 CHT	Implement <code>f1ip-normal</code> operation adapter-wide inside 'Coupl...	2 months ago	 Allwmake	change the log in allwmake	last week
 FF	Implement <code>f1ip-normal</code> operation adapter-wide inside 'Coupl...	2 months ago	 CHANGELOG.md	Changelog for v1.3.1	2 years ago
 FSI	Implement <code>f1ip-normal</code> operation adapter-wide inside 'Coupl...	2 months ago	 CITATION.cff	fill correct info	3 years ago
 Make	update options file to compile with cosimio	last week	 CODE_OF_CONDUCT.md	Update master (v1.0.0) (precice#177)	5 years ago
 changelog-entries	Update meshConnectivity claims (precice#393)	2 months ago	 CONTRIBUTING.md	Document running actions locally and update custom build C...	last year
 docs	Generic Read Write module (precice#346)	2 months ago	 CouplingDataUser.C	Implement <code>f1ip-normal</code> operation adapter-wide inside 'Coupl...	2 months ago
 module-generic	Implement <code>f1ip-normal</code> operation adapter-wide inside 'Coupl...	2 months ago	 CouplingDataUser.H	Implement <code>f1ip-normal</code> operation adapter-wide inside 'Coupl...	2 months ago
 tools	Update create-release-archives.sh (fix typo in naming)	2 years ago	 Interface.C	Implement <code>f1ip-normal</code> operation adapter-wide inside 'Coupl...	2 months ago
 .clang-format	Bump versions to OpenFOAM v2512 and preCICE v3.3.0 (prec...)	3 months ago	 Interface.H	Implement <code>f1ip-normal</code> operation adapter-wide inside 'Coupl...	2 months ago
 .gitattributes	Add a .gitattributes for correct language recognition from Git...	9 years ago	 LICENSE	Add license	9 years ago
 .gitignore	Adds functionality to allow FSI to work with solids4foam as th...	4 years ago	 README.md	Bump versions to OpenFOAM v2512 and preCICE v3.3.0 (prec...)	3 months ago
 .markdown-link-check-config.json	Update master (v1.0.0) (precice#177)	5 years ago	 Utilities.C	Allow for case-insensitive exchanged field names (precice#391)	2 months ago
 .markdownlint.json	Markdown-lint: Disable MD034 (precice#290)	3 years ago	 Utilities.H	Allow for case-insensitive exchanged field names (precice#391)	2 months ago
 .pre-commit-config.yaml	Add clang-format pre-commit hook (precice#331)	2 years ago	 preciceAdapterFunctionObject.C	Rectify version specifications for OPENFOAM and OPENFOA...	3 years ago
 Adapter.C	Fix dangling reference (precice#401)	last month	 preciceAdapterFunctionObject.H	Add timers (precice#256)	4 years ago
 Adapter.H	Change NULL pointer values to C++11 nullptr (precice#396)	2 months ago			

- **CoSimIOAdapterFunctionObject.c/.h**: Implements the OpenFOAM function object that initializes, drives, and finalizes the coupling with CoSimIO during the simulation.
- **Adapter.c/.h**: Contains the code itself. Managing time-stepping, data exchange, and interaction with CoSimIO.
- **Interface.c/.h**: Provides a wrapper around the CoSimIO, exposing simplified functions for reading, writing, and advancing the coupling.

Developed Approach: Strong-Coupling Adapter



Developed Approach: Strong-Coupling Adapter

Checkpointing

```
void preciceAdapter::Adapter::WriteMeshCheckpoint()
{
    DEBUG(adapterInfo("Writing a mesh checkpoint..."));

    // Store all the fields of type mesh surfaceScalar (phi)
    for (uint i = 0; i < mMeshSurfaceScalarFields.size(); i++)
    {
        *(mMeshSurfaceScalarFieldCopies.at(i)) == *(mMeshSurfaceScalarFields.at(i));
    }

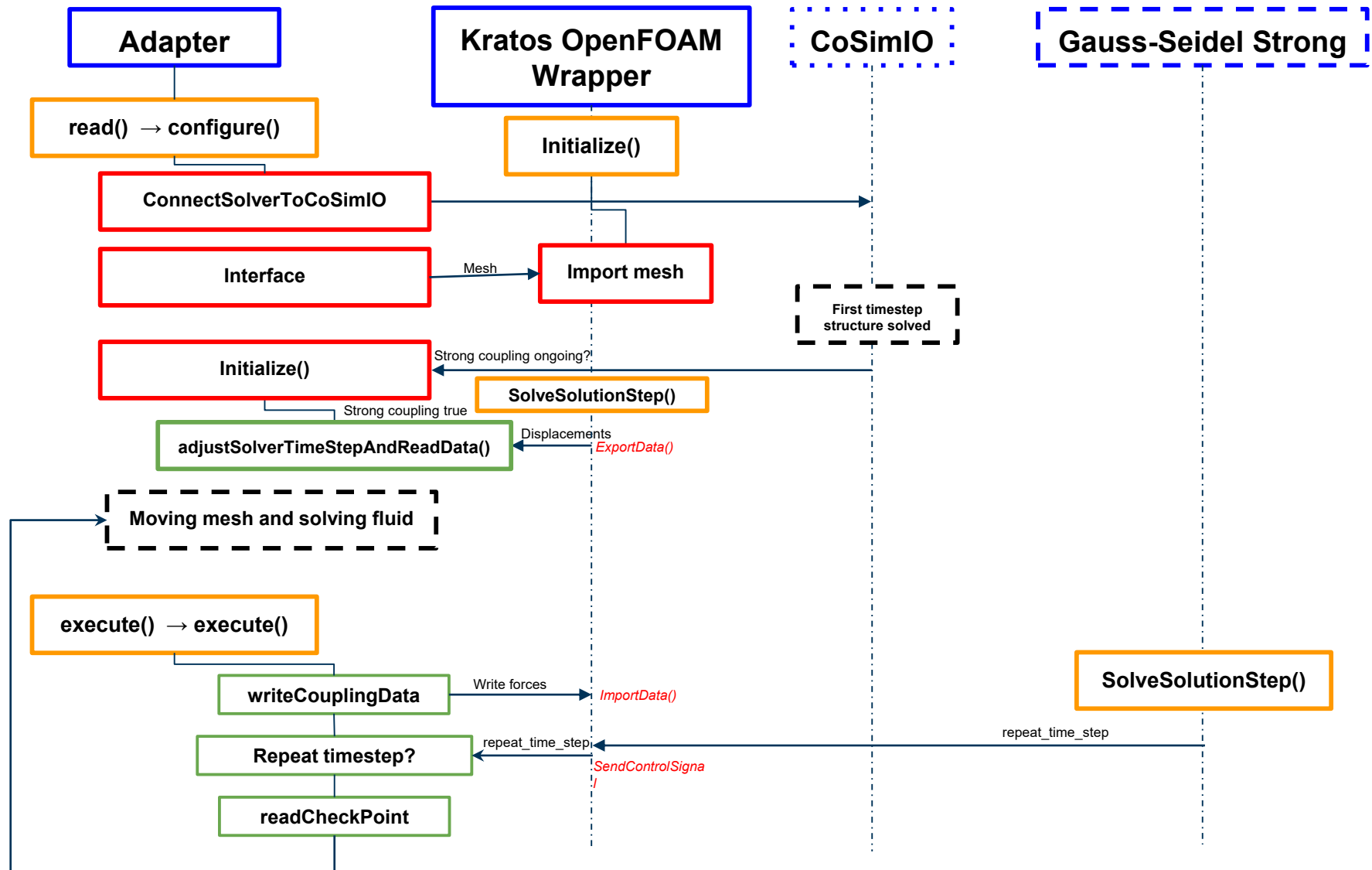
    DEBUG(adapterInfo("Storing mesh points..."));

    // Store mesh points
    // swap pointers
    *(mMeshOldPoints) = *(mMeshPoints);
    *(mMeshPoints) = mMesh.points();

    DEBUG(adapterInfo("Mesh checkpoint for time t = " + std::to_string(mRunTime.value()) + " was stored."));

    return;
}
```

Developed Approach: Strong-Coupling Adapter



Developed Approach: Strong-Coupling Adapter

```
libs ("libCoSimIOAdapterFunctionObject.so");
functions
{
    CoSimIO_Adapter
    {
        type CoSimIOAdapterFunctionObject;
        errors strict; // Available since OpenFOAM v2012
    }
}
```

 blockMeshDict

 controlDict

 CoSimIODict

 decomposeParDict

 fvSchemes

 fvSolution

 fvSolution_new

```
FoamFile
{
    version      2.0;
    format        ascii;
    class         dictionary;
    object        CoSimIODict;
}
interfaces
{
    interface_flap
    {
        mesh            interface_flap;
        patches          (Mok);
        locations        FaceCenters;
    }

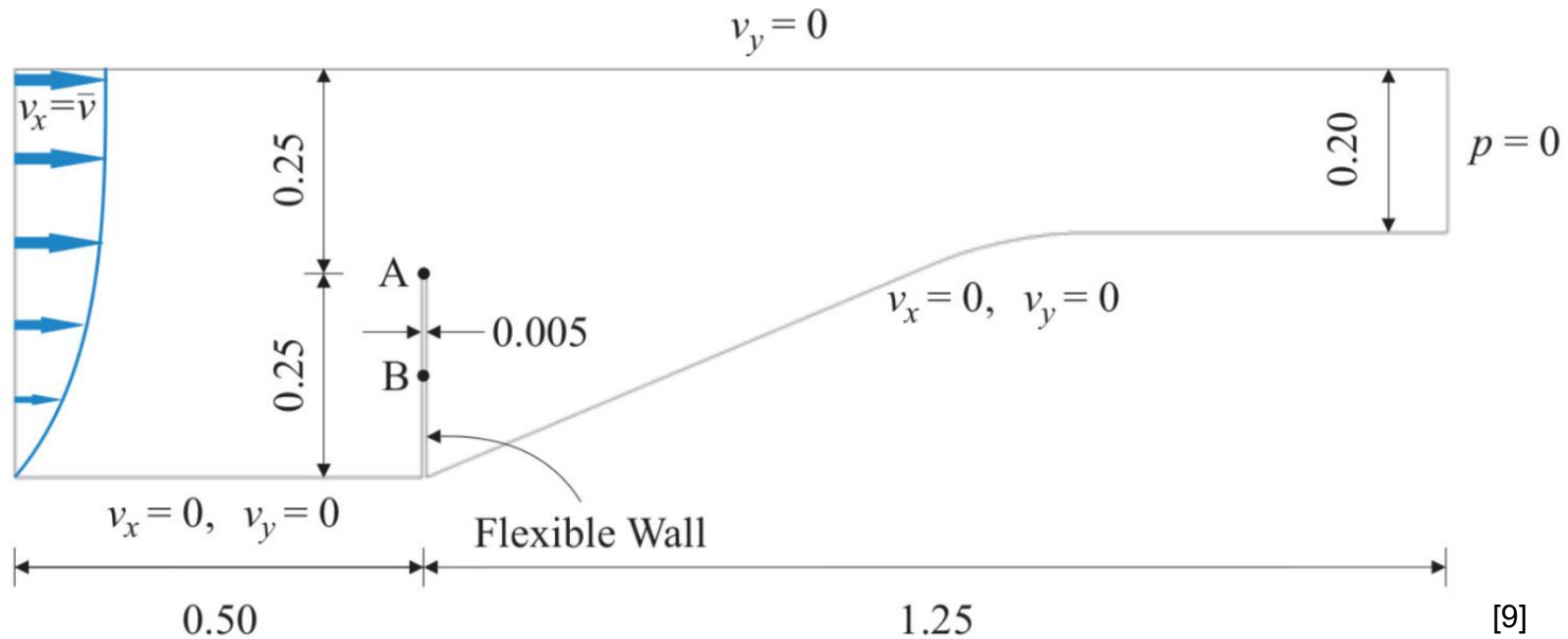
    ReadData
    (
        | | Displacement-Flap
    );

    WriteData
    (
        | | Force-Flap
    );
};

FSI
{
    rho rho [1 -3 0 0 0 0 0] 956.0;
}
```

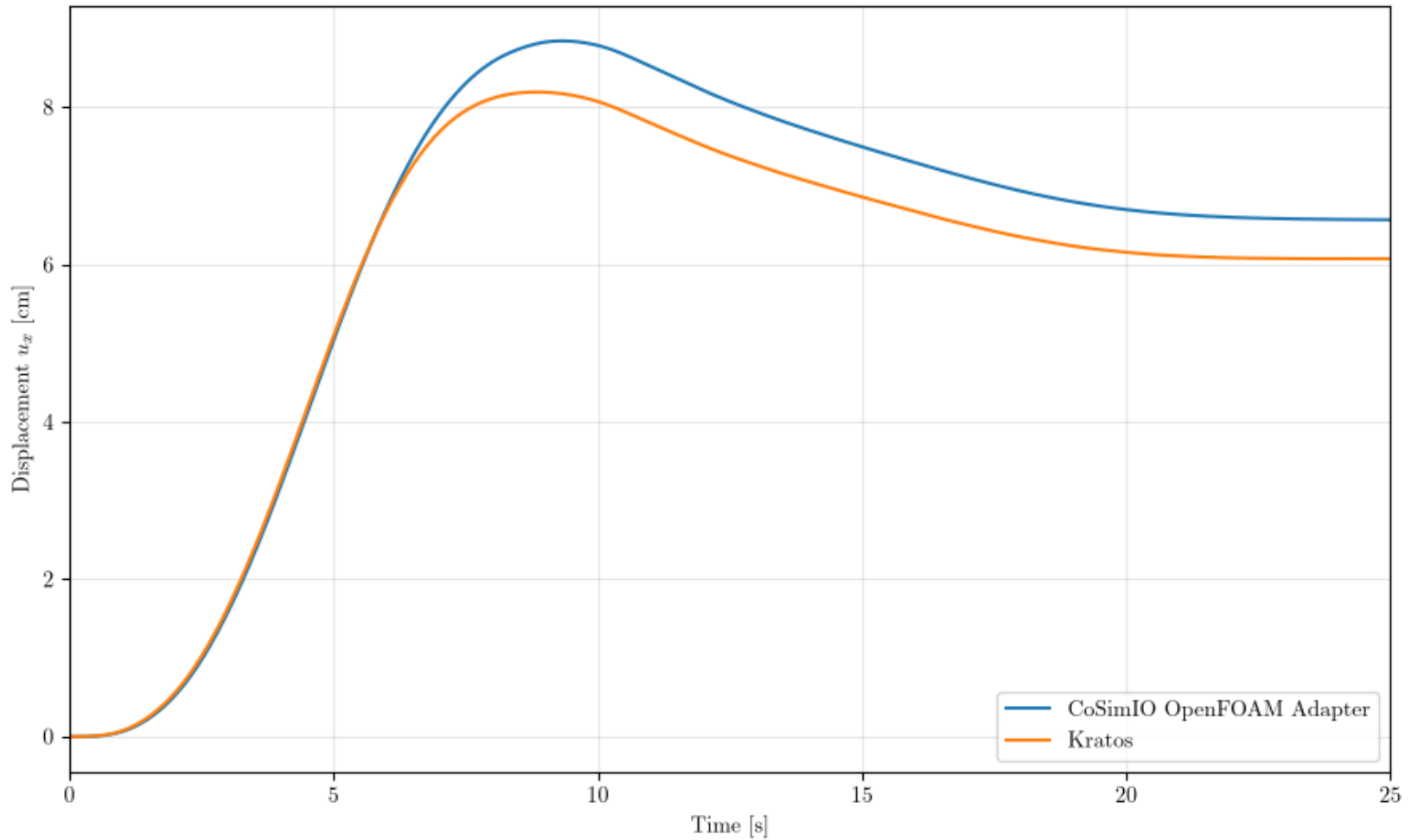
Benchmark Cases

Benchmark Cases: Mok

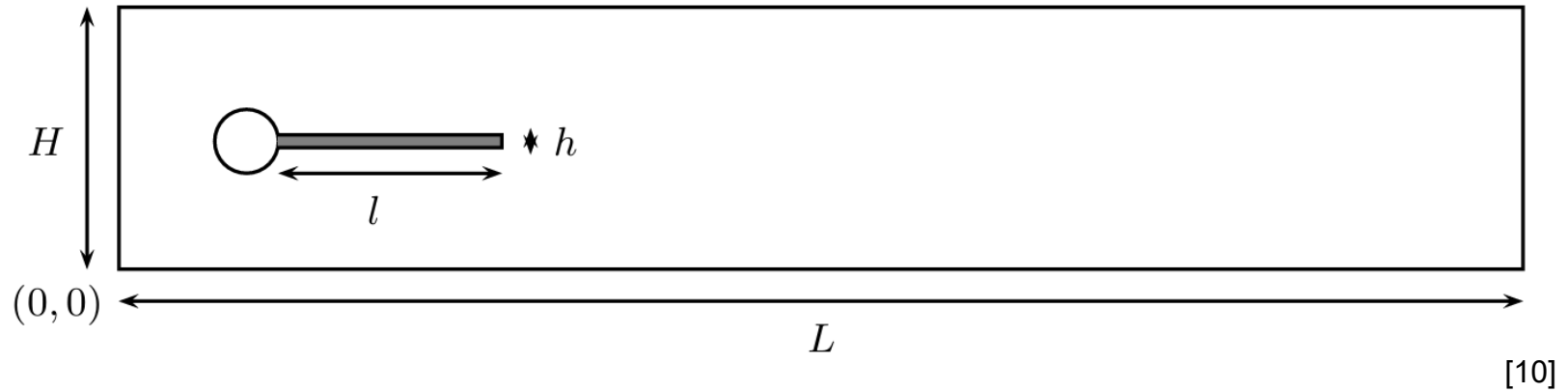


Parameter	Kratos Benchmark	OpenFOAM-Kratos Adapter
Fluid mesh nodes	3,590	3,886
Structure mesh nodes	505	505
Time step (s)	0.1	0.05
Start–end time (s)	0–25	0–25

Benchmark Cases: Mok

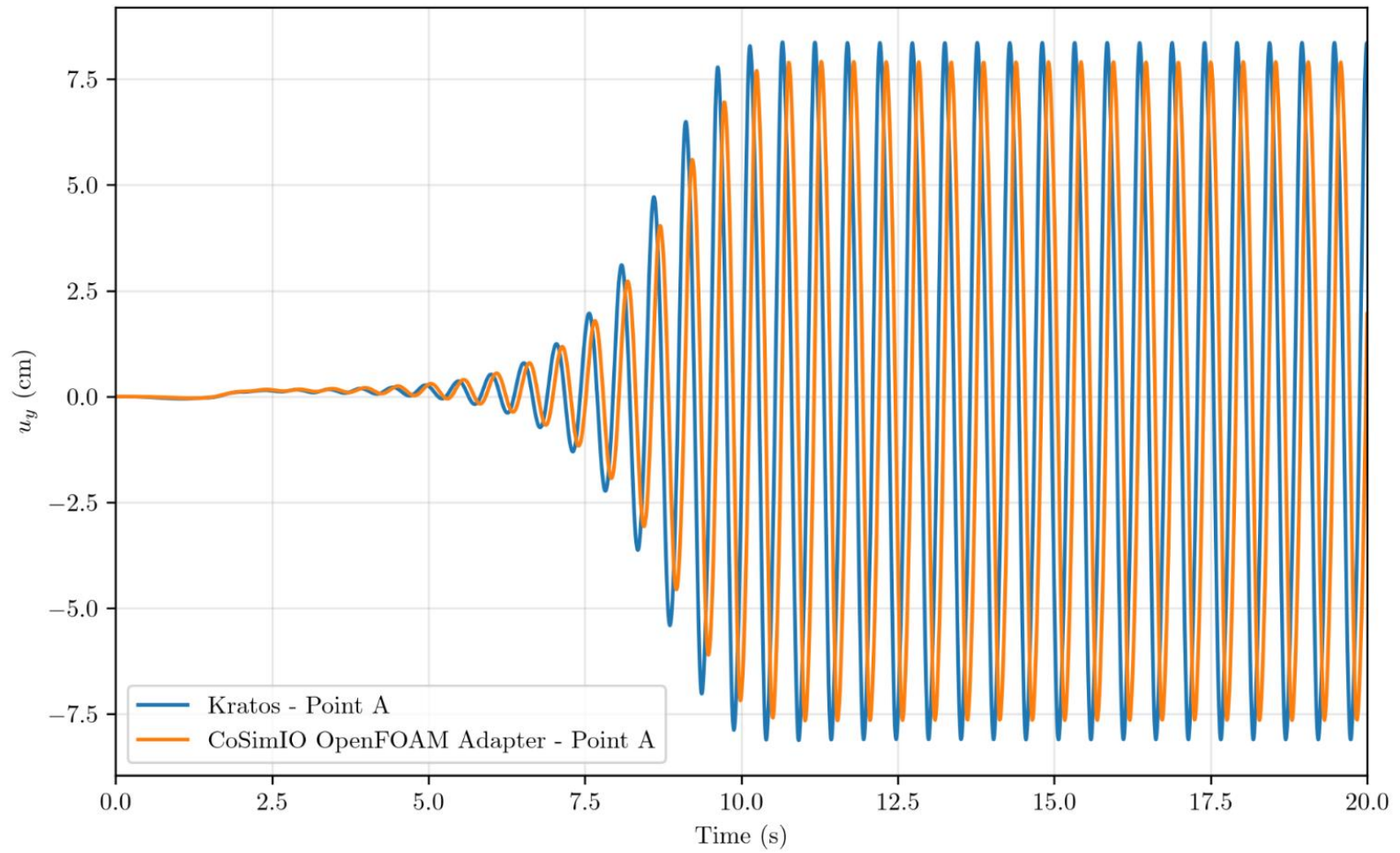


Benchmark Cases: Turek

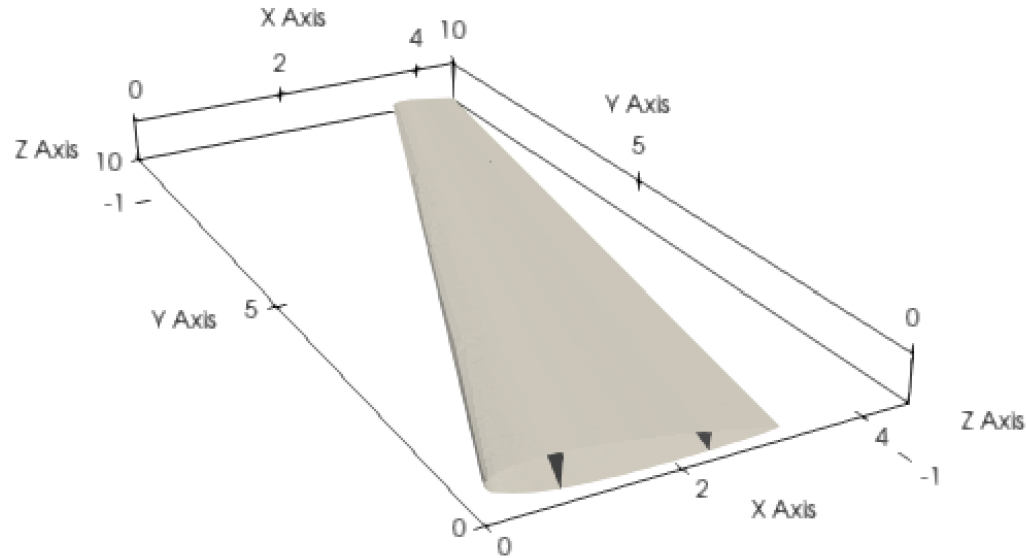


Parameter	Kratos Benchmark	OpenFOAM-Kratos Adapter
Fluid mesh nodes	15,577	13,120
Structure mesh nodes	2,816	2,816
Time step (s)	0.002	0.002
Start–end time (s)	0–20	0–20

Benchmark Cases: Turek

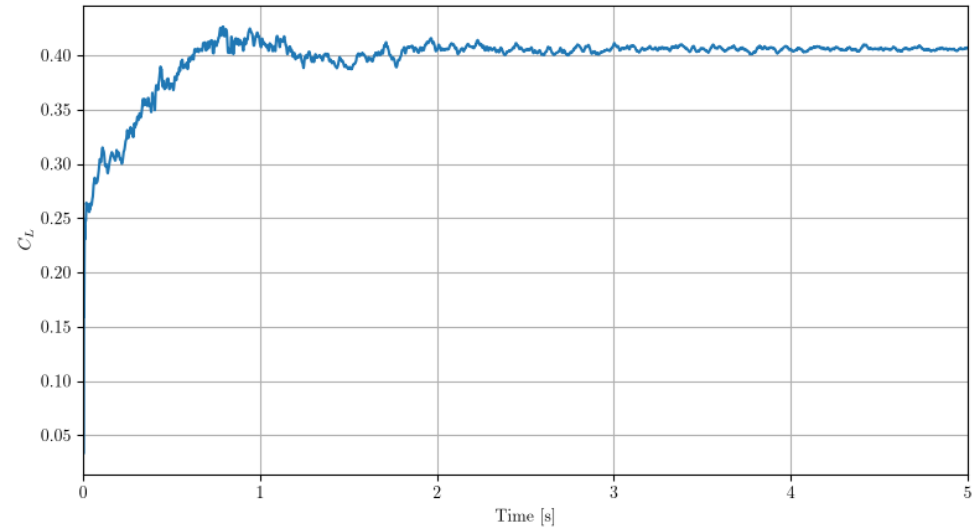
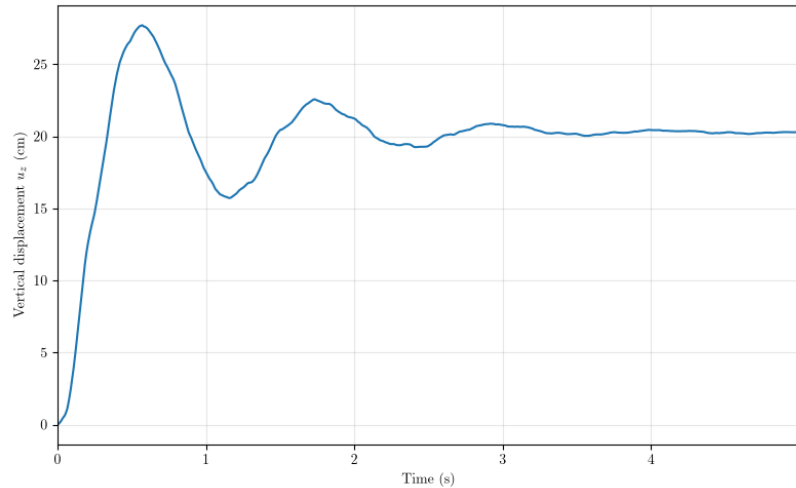


Benchmark Cases: 3D Wing



Parameter	OpenFOAM-Kratos Adapter
Fluid mesh nodes	87,782
Structural mesh nodes	5,326
Time step (s)	0.001
Start–end time (s)	0–5

Benchmark Cases: 3D Wing



Note: A rough estimate of the wing's rigid lift coefficient (C_L) can be obtained using the following approximation.

$$C_L = C_{L\alpha}(\alpha - \alpha_{L0}) = C_{l\alpha} \frac{A}{2 + \sqrt{4 + A^2(1 + \tan^2(\Lambda_{25}))}} (\alpha - \alpha_{L0})$$

For this wing, considering a NACA 0012 symmetric airfoil:

$$A = \frac{b^2}{S} = 10 \quad \Lambda_{25} \approx 17.22^\circ \quad \rightarrow \quad C_L(5^\circ) \approx 0.396$$

$$C_{l\alpha} \approx 5.73 \text{ 1/rad}$$

Outlook

- **OpenFOAM–Kratos adapter** developed for weak and strong FSI coupling through CoSimIO.
- Validated on **Mok** and **Turek** benchmarks, and applied to a **3D turbulent wing** case with $\sim 2.8\%$ error in lift coefficient.
- **Future work**: mesh refinement and **MPI**-based parallelism for larger FSI simulations.

Thanks for your attention!

References

- [1] H. G.Weller et al. “A tensorial approach to computational continuum mechanics using objectoriented techniques”. In: Computers in Physics 12.6 (1998), pp. 620–631.doi:10.1063/1.168744.
- [2] Pooyan Dadvand, Riccardo Rossi, and Eugenio Oñate. “An Object-oriented Environment for Developing Finite Element Codes for Multi-disciplinary Applications”. In: Archives of Computational Methods in Engineering 17 (2010), pp. 253–297. doi: 10.1007/ s11831 - 010-9045-2.
- [3] Pooyan Dadvand et al. “Migration of a generic multi-physics framework to HPC environments”. In: Computers & Fluids 80 (2013), pp. 301–309.doi:10.1016/j.compfluid.2012.02.004.
- [4] Vicente Mataix Ferrándiz et al. KratosMultiphysics/Kratos: Release v10.2.3. Version v10.2.3. 2025. doi: 10 . 5281 / zenodo . 15687676. url: <https://doi.org/10.5281/zenodo.15687676>.
- [5] Gerasimos Chourdakis. “A General OpenFOAM Adapter for the Coupling Library preCICE”. Master’s Thesis. München: Technische Universität München, Oct. 2017.
- [6] Ashish Shankar Darekar. “Development of a General OpenFOAM-Coupling-Adapter for the KratosCoSimulation Multiphysics Solver”. Master’s Thesis. Technical University of Munich, 2021.

References (2)

- [7] Gerasimos Chourdakis et al. “preCICE v2: A sustainable and user-friendly coupling library”. In: Open Research Europe 2 (2022), p. 51.
- [8] Gerasimos Chourdakis, David Schneider, and Benjamin Uekermann. “OpenFOAM-preCICE: Coupling OpenFOAM with External Solvers for Multi-Physics Simulations”. In: OpenFOAM Journal 3 (2023), pp. 1–25. issn: 2753-8168. doi: 10.51560/ofj.v3.88.
- [9] Stefan Turek and Jaroslav Hron. “Proposal for Numerical Benchmarking of Fluid-Structure Interaction between an Elastic Object and Laminar Incompressible Flow”. In: Fluid-Structure Interaction. Springer, 2006, pp. 371–385.
- [10] Rubén Zorrilla. Mok Benchmark. https://github.com/KratosMultiphysics/Examples/blob/master/fluid_structure_interaction/validation/fsi_mok/README.md. KratosMultiphysicsExamples, accessed July 2026.